

New Agriculture Wealth Opportunities-Driving Horticulture and Agribusiness Networking (NAWO-DHAN)

1. Introduction

Kerala state has a total cropped area of 2523014.31ha of which more than 85% of the area is covered under perennial crops and only 15% of the area is utilised for food crops. It is estimated that there is 103334 ha of fallow lands available in the State of which 53914 ha is current fallow and 49420 ha is permanent fallow land use 2021-2022 ecostat).

The State faces food shortage and produces less than 15% of its food requirement especially rice. Current vegetable requirement of Kerala is 30 lakh MT per year and the production is 17.24 lakh MT per year. There is gap of 13.76 lakh MT to be met from the neighbouring states. Similarly the current fruit requirement in the State is 40 lakh MT/year whereas the annual production is only around 26 lakh MT/year. The challenges faced by State in Agriculture are numerous. The labour wages in Kerala is 2.5 times more than the National wage rate. 96% of operational holdings are under marginal categories. There is a reduction of gross crop area from 29, 33,000 ha in 1970-71 to 25, 23,000 ha in 2021-22 (14 % reduction). Unplanned crop production, lack of adequate post harvesting technologies, price instability, climate changes etc. are also add to the challenges faced by State Agriculture.

Food and beverage markets in India is forecasted to grow at a CAGR (Cumulative Agriculture Growth Rate) of 9.94 % from 2022-27 is a positive sign for creating opportunities in agriculture. Kerala also received GI tag for 24 crops as well as top producer of Pepper, Cardamom, Coconut, Tea, Jack fruit, Tapioca, nutmeg, cocoa etc. By 2030, Indian annual household consumption is expected to triple and making India the fifth largest consumer in world. The potential for food export is also going in an upward trend and logistics advantages can also make Kerala as one of the potential States in export. Convergence with different State and Central scheme for planned agriculture production activities may utilise to achieve the benefits.

Unorganised farmers are to be organised through formation of FPOs, Krishikoottams, SHGs, JLGs etc. as it is the only way of institutionalisation and attaining an organised farming practice. Implementation of innovative farm solutions, R&D on agriculture

activities, efficient logistics, value chain acceleration, food supply chain traceability & solutions, branding & marketing initiatives are the immediate needs of the farming sector.

There is a high need to increase our food crop production area for which the present fallow and intercropping area should be well utilised in a phased manner. Most of the land are kept fallow or unutilised due to the scarcity of resources, labour shortage, less interest in cultivation as it is not remunerative, inability of the land owner to look into the daily activities of his crops as they are aged or NRIs or employed elsewhere. Some good initiatives are already taken in line with the objectives of the proposal in the state owned Aralam Farm at Kannur District where land is made available to the cultivators and service farming is done by the farmer groups successfully in the past 2 years. In order to demonstrate, that this type of farming have prospects throughout the state, the same can be replicated in the unutilised land available in the unutilised /underutilised Government owned lands.

Since the income from agriculture is not taxable is one of the attractions to invite more people/MNCs to invest in Hi-tech value cropping through **NAWO-DHAN** proposal. KABCO (Kerala Agro Business Company) will act as the SPV (Special Purpose Vehicle) for carrying out the **NAWO-DHAN** project. KABCO will identify the fallow land available with all Govt. own land, PSUs, other institutes etc. and act as a land pooling bank for developing land to hi-tech agri. farms through NAWO-DHAN proposal.

NAWO-DHAN may be started on a pilot basis inviting EoI from those who are willing to make available their unutilised /underutilised/fallow lands for farming. The availability of fallow land, intercropped area, suitable climate, soil fertility status, water & input availability and logistics are different criteria used for the selecting special farms. The land can be apportioned as blocks of 10 acre and can be made available to the farmers for cultivation for high-tech value cropping. The successful frame work model generated through this high-tech value cropping can be replicated in other Government and private owned lands.

The cost of cultivation is high due to the high labour rate persistent in our State. So the profitability of farmer depend up on the selection of high value crop as well as reduce the cost of production by adopting smart agriculture.

2. The main objectives of the programme are :

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- Identification of fallow land in Govt owned land, PSUs etc and pooling the land to inventory bank operated by KABCO.
- To empower the farmer to provide land and input on charges.
- Techno-park model to be adopted by providing land for cultivation as units of 10 acre each.
- Infrastructure like water, electricity to be provided on cost basis.
- Possibility of tying up with other national/International agencies involved in hi-tech farming for exchange of technology can be explored.
- To increase the area of food/vegetable/fruit crops in fallow land /intercropping cultivation available in the identified places.
- The selection of crops for cultivation at the assigned units will be decided by the farmer.
- Expecting an increase in the average production of 30 lakh MT per year quantity of vegetable and fruit crops.
- Increase the productivity of existing sole crops by 40 % by utilising the intercrop area for farming.
- To create a frame work for the high-tech value cropping to generate more income to the farm.
- Demonstrate the new technology and farming technique to farming community.
- Overall development of the assigned area.
- Creation of employment opportunities to the landless unemployed section of the society.
- Attracting youth /new players towards agriculture.
- Economic upliftment of the community nearby.

3. The main programmes of the project are the following:

- a. Selection of available cultivable land through EOI procedure by KABCO.
- b. Selection of farmer/farmer group
- c. Handholding for availing credit facilities
- d. Providing the inputs and machineries available at Farm

- e. Providing crop insurance support
- f. Providing technical guidance
- g. Possibility of convergence with Schemes of Department of Agriculture

a. Selection of available cultivable land.

It is estimated that 9000 ha of unutilised land is available in the state for intercropping. The available cultivable land has to be brought under high-tech value cropping in a phased manner. The procedure can be a replicate model of Techno Park, by demarking 10 acres as one unit (maximum contiguous area). The available unutilised land can be divided into multiples of 10 acre units. Farmer can be invited to take up the 10 acre as one unit for high-tech value intercropping and the land charges can be fixed for one unit through a tender procedure. Farmer can avail the infrastructure facilities like irrigation water, electricity, machineries, farm roads, store house etc. if available in the location based on prefixed service charges.

b. Selection of Farmer

Scarcity of cultivable land is one of the reasons that hinders the farmers/Farmer group to adopt farming in large areas. There are farmers as well as farmer groups like FPCs, Start-ups Krishikootams, Kudumbasree, Janashree, farmers clubs, etc who are willing to undertake farming if cultivable land is available. Individuals or farmer groups may be selected through a tender (QCBS method) procedure and the identified land may be provided for cultivation for a period of 1 year to undertake high-tech value intercropping. The QCBS document will elaborate detailed criteria for technical and financial bid with 60: 40 weightage.

c. Handholding for availing credit facilities

In order to initiate cultivation in assigned land more capital investment is required for land preparation, adoption of new technology, cultivation of high value crop etc and hence cost of cultivation of the crops raised will be higher than that of normal crops. The persons or agency proposed to cultivate may not have any resources to mobilize the cost of cultivation. Hence credit facilities can be familiarised to the cultivators to avail better option. The primary co-operative societies and other financial institution can play a major role to extend credit facilities to the cultivators. If any direction is needed on the ownership of land for availing credit facilities, Government may support with suitable mechanism. As the *New Agriculture Wealth Opportunities-Driving Horticulture and Agribusiness Networking (NAWO-DHAN)*

cultivators are not having the ownership of the land, special attention may be given from the Government to make available the credit facilities to the cultivators.

d. Providing the inputs and machineries available at Farm

Suitable machineries and implements are required for the preparation of land for cultivation. The input such as seeds and planting materials, manures, bio pesticides, farm machineries etc which are available at farms/ Agro Service centres will be provided as per prevailing rate. Details regarding the services offered by Karshika Karma Senas / Agro Service Centres nearby areas may be provided to the entrepreneur.

e. Providing crop insurance support

Crop failures due to various reasons such as occurrence of natural calamities, outbreak of pests and diseases etc. are common in agriculture. The cultivators should be properly compensated in case of crop failure and for that crop insurance facilities may be extended to these cultivators also.

f. Providing technical guidance

By providing proper technical guidance, the farming can be made profitable. The Technical Officers of the Farms/ Krishi Bhavans/KAU will provide timely technical guidance throughout the cropping period. The following are the technical guidance offered

- To ensure supply of quality inputs like seeds, fertilizers and pesticides
- Proper and timely irrigation facilities for carrying out farming operations
- To check the past pest and disease attack and to take remedial action to overcome them
- Agricultural information dissemination
- Farming operations to be calendared so as to ensure maximum and sustainable use of natural resources
- Intervention in Location specific farming techniques
- Mobilising institutional credit for farming activities
- Providing crop insurance coverage
- Promotion of mechanization activities for farming

- Promotion of eco-friendly practices in agriculture

g. Possibility of convergence with Schemes of Department of Agriculture

The possibility of integrating with different schemes can be explored for the benefit of farmers who undertake high-tech value intercropping in Farms.

4. Operational guidelines

Frame work for NAWO-DHAN

Release of EOI for Identification of Suitable Area for Hitech-Value cropping (based on land availability, infrastructure and logistics) by KABCO



Enlisting and Prioritising Suitable land for Hitech-Value cropping (Land area, intercrop/pure crop/fertility status/Resources availability) –KABCO



Call for proposal by QCBS tender procedure

(Preparation of TOR and RFP for identifying suitable entrepreneurs/Farmers/Pvt Players/FPOs/Start ups/Farmer Groups etc and e tendering)-KABCO



Empanelling the entrepreneurs/farmers/start ups



Awarding contract after executing proper agreement



Assuring the farming by providing resources like water/electricity/inputs/machinery etc. on rental /cost basis .



Monitoring of harvesting to assure proper remittance of charges

(A fixed amount may be realised as charges for the land or a percentage of the profit generated from farming may be taken as charges by the land owner)

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5. Global Scenario

Farm on rent is a way in EU countries to enhance the production and productivities of food crops. Sustaining the agriculture is the motive behind the renting out of land to tenants and providing all the support to enhance the production of food crops.

6. Implementing Agency

The project will be implemented by the SPV, KABCO under the direct supervision of the Department of Agriculture.

7. Conclusion

The **NAWO-DHAN** is one of the milestones proposed for utilising the available land and resources at our farms to generate sustainable income by increasing production and proper utilisation of resources. The pilot project proposed can be replicated in other unutilised public lands for the betterment of farming community. The younger generation can be attracted to the field of farming with attractive package framed through rental farming. Start-ups in Agriculture can play a major role for building future Kerala by adopting this type of farming by renting the land.